

COSC 301 Report

Keeran Naidu, Ali Afoud, Mandira Samarasekara

April 6, 2026

Questions

Which Toronto neighbourhoods pose the greatest safety risk and does crime severity correlate with traffic collision severity at the neighbourhood level? Where and when crimes are likely to happen using predictive models?

1 Introduction

This project leverages the Toronto Police Service’s publicly available open data initiative to conduct a comprehensive spatio-temporal analysis of criminal activity across the city. The dataset spans from 2014 to the present date and provides granular, event-level records of reported incidents. By transforming this raw data into actionable intelligence, the project seeks to uncover patterns, forecast risk, and support data-driven community safety decisions.

Primary Analytical Goals:

1. Develop an integrated, functional modeling pipeline that predicts *when* and *where* specific crime types are statistically more likely to occur, outputting predictions with interpretable confidence levels.
2. Compute a dynamic, transparent Neighborhood Safety Score to rank areas and visualize safety trends over time.
3. Exploratorily investigate correlations between crime profiles and socioeconomic indicators while explicitly acknowledging the limitations and ethical considerations of such inference.

We thus created a system that allows citizens and policymakers to better understand the distribution of crime and traffic incidents across the city of Toronto.

2 Method

We used two major datasets from Toronto’s Police Service Open Data:

- **Major Crime Indicators** (<https://open.toronto.ca/dataset/major-crime-indicators/>): Contains reported crimes across Toronto categorised into 6 major crime types. We examined `EVENT_UNIQUE_ID`, `OCC_DATE`, `NEIGHBOURHOOD_158`, `CSI_CATEGORY`, `OFFENCE`, `DEATH`, `INJURIES`, `EVENT_TYPE`, and `PREMISE_TYPE`.

- **Motor Vehicle Collisions involving Killed or Seriously Injured Persons** (<https://open.toronto.ca/dataset/motor-vehicle-collisions-involving-killed-or-seriously-injured-persons/>): Contains traffic collision records with features including `EVENT_UNIQUE_ID`, `OCC_DATE`, `FATALITIES`, `INJURY_COLLISIONS`, `AUTOMOBILE`, `MOTORCYCLE`, `PASSENGER`, `BICYCLE`, `PEDESTRIAN`, `NEIGHBOURHOOD_158`, `FTR_COLLISIONS`, and `PD_COLLISIONS`.

We used these datasets to rank neighbourhood safety and build predictive models for where and when incidents were likely to occur. We downloaded the datasets as CSVs and loaded them into a PostgreSQL database using Python. Both cleaned datasets were stored in PostgreSQL database with typed schemas. Safety scores were computed via SQL aggregation queries and exported as CSVs so they can be used in Power BI. For our tech stack, we used Python for scripting and feature engineering, Power BI for data visualization, and PostgreSQL for data storage.

2.1 Data Cleaning Pipeline

Before analysis, the raw crime data underwent a rigorous multi-step cleaning process:

- **Initial Filtering:** Any records prior to 2014 were removed. Rows where the neighbourhood was listed as “NSA” (Non-Standard Area) or null, or where the occurrence date was null, were also discarded.
- **Event Type Reconciliation:** A script scanned all data and grouped rows by `EVENT_UNIQUE_ID`. It identified the correct `EVENT_TYPE` from rows that had it (even if `CSI_CATEGORY` was missing). Orphaned rows (with `EVENT_TYPE` but no `CSI_CATEGORY`) were deleted as incomplete records. For rows that had `CSI_CATEGORY` but no `EVENT_TYPE`, the `EVENT_TYPE` was filled in from other rows with the same `EVENT_UNIQUE_ID`.
- **Deduplication:** An MD5 hash was created for each row using all columns. Only the first occurrence of each unique hash was kept, removing all subsequent duplicate rows after user confirmation.
- **Feature Enrichment:** Time-based features were added to the remaining data, including `week_day` (e.g., “Monday”), `season` (meteorological seasons), and `holiday` (a boolean indicating if the date is a Canadian or Ontario holiday).

Traffic data was loaded directly after similar temporal feature engineering, with both tables loaded row by row with date parsing and type conversion handled in Python before insertion.

2.2 Safety Score Calculation

For both datasets, we created safety scores to rank neighbourhood safety. To improve interpretability, all scores were normalized across years and neighbourhoods onto a 0-100 scale (higher = safer).

2.2.1 Crime Safety Score

For the crime data, we calculate a weighted sum across all years for each crime event. The recency weight gives more importance to recent years (2025) compared to older years (2014). The core calculation for each crime event is:

$$\text{Event Weight} = \text{CSI_WEIGHT} \times \text{OFFENCE_WEIGHT} \times \sum_{y=2014}^{2025} (\text{RECENCY_WEIGHT}_y \times \text{Frequency}_y)$$

Where:

- **CSI_WEIGHT**: Assigned based on the CSI category (e.g., Assault = 1.0, Non-MCI = 0.1)
- **OFFENCE_WEIGHT**: Fine-grained weight for specific offences within a CSI category (e.g., Aggravated Assault higher than Common Assault)
- **RECENCY_WEIGHT_y**: Linear decay from 1.0 (2025) to 0.2 (2014), giving recent years greater influence
- **Frequency_y**: Number of times this specific offence occurred in neighbourhood n during year y

For each neighbourhood, we sum the Event Weight across all crime events, then normalize to a 0-100 safety score:

$$\text{Crime Safety Score}_n = 100 \times \left(1 - \frac{\text{Raw Danger Score}_n - \min(\text{Raw Danger Score})}{\max(\text{Raw Danger Score}) - \min(\text{Raw Danger Score})} \right)$$

Where Raw Danger Score _{n} is the total weighted sum of all events in neighbourhood n . This formulation ensures that neighbourhoods with more frequent and severe crimes receive lower safety scores. Finally, we ranked all 158 neighbourhoods in descending order from safest to most dangerous.

2.2.2 Traffic Safety Score

The traffic safety score follows a similar recency-weighted approach. For each neighbourhood, we first apply recency weights to collision counts, then calculate rates per 1000 recency-weighted collisions for five collision types:

$$\text{Rate}_t = \frac{\sum_y (\text{RECENCY_WEIGHT}_y \times \text{Collisions}_{t,y})}{\sum_y (\text{RECENCY_WEIGHT}_y \times \text{Total Collisions}_y)} \times 1000$$

Where $t \in \{\text{fatalities, injury, fail-to-remain (FTR), vulnerable road users, property damage (PD)}\}$.

The raw danger score combines these rates with severity weights (fatalities weighted highest at 5, property damage lowest at 1):

$$\text{Raw Danger}_n = 5 \times \text{Rate}_{\text{fatal}} + 3 \times \text{Rate}_{\text{injury}} + 2 \times \text{Rate}_{\text{FTR}} + 2 \times \text{Rate}_{\text{vulnerable}} + 1 \times \text{Rate}_{\text{PD}}$$

To account for collision volume (neighbourhoods with many total collisions should be penalized even if per-collision rates are moderate), we blend normalized danger score (70%) with normalized collision volume (30%):

$$\text{Blended Danger}_n = 0.7 \times \text{norm}(\text{Raw Danger}_n) + 0.3 \times \text{norm}(\text{Total Collisions}_n)$$

Finally, we invert to create a safety index (higher = safer) and scale to 0-100:

$$\text{Traffic Safety Score}_n = 100 \times (1 - \text{Blended Danger}_n)$$

This produces comparable 0-100 safety scores across all 158 Toronto neighbourhoods.

2.3 Power BI Implementation

Both safety score calculations were successfully implemented in Power BI, enabling interactive exploration of Toronto’s neighbourhood safety landscape. The implementation features:

- **Dynamic Heatmaps:** Toronto geo-map visualizations where neighbourhoods are coloured from green (safest) to red (most dangerous), updating in real-time based on user filters.
- **Multi-dimensional Filtering:** Users can filter by:
 - Time range (sliding window from 2014-2025)
 - Crime CSI category (Assault, Auto Theft, Break & Enter, etc.)
 - Specific offence sub-types (e.g., within Assault: Common Assault, Aggravated Assault, Assault with Weapon)
 - Collision victim types (Pedestrian, Bicyclist, Motorcyclist, Automobile Passenger)
 - Collision severity (Fatal, Injury, Property Damage Only, Fail-to-Remain)
- **Neighbourhood Drill-Down:** Clicking any neighbourhood reveals:
 - Detailed safety score breakdown by crime type and year
 - Temporal trend lines for both crime and traffic incidents
 - Ranking comparison against city average and top/bottom 10 neighbourhoods
- **Cross-Dataset Correlation View:** A scatter plot comparing crime safety score vs. traffic safety score for all neighbourhoods, with ability to select individual neighbourhoods for deeper analysis.

This Power BI dashboard provides stakeholders from city planners to community safety organizations with an intuitive, data-driven tool to understand and visualize safety risks across Toronto.

2.4 Predictive Models

Beyond safety scores, we created four predictive models for **traffic collisions**, **assault**, **auto theft**, and **break and enter**. These models predict whether a crime type would occur in a neighbourhood within a fixed time window (3 days for collisions and assault, 7 days for auto theft and break and enter) for all neighbourhoods. Models were trained on 2014-2024 data and validated on 2025 data.

The predictive models focused only on these four categories as they contained the most observations. (What a safe city Toronto is other crime tables contained data too sparse to train a reliable model!)

The modeling approach evolved significantly. We initially started with Random Forest feature elimination but then shifted to XGBoost based on methodologies found in research papers. To combat bias towards positive predictions, we experimented with LightGBM and CatBoost. LightGBM proved most effective in maintaining balance between positive and negative predictions, lowering the number of false negatives and false positives. This was followed by extensive hyperparameter tuning, allowing for rigorous experimentation.

Crucially, our model operates on a **3-day window** instead of the 2-week windows common in research papers. We found that 1-week and 2-week windows were too large they virtually guaranteed that *something* would happen, making the model trivially predict “true” for everything. Reducing the window to 3 days created a more meaningful and challenging prediction task.

2.4.1 Model Interpretability with SHAP

Given that crime predictions are inherently sensitive and can influence decisions about resource allocation, public safety, and community trust, we recognized that deploying a pure “black box” model would be insufficient. Stakeholders need to understand *why* a model makes a particular prediction to trust and act upon it.

To address this, we integrated **SHAP (SHapley Additive exPlanations)** into our modeling pipeline. SHAP provides a unified framework for explaining individual predictions by quantifying the contribution of each feature to the model’s output. For any given prediction e.g., “Neighbourhood X has a 73% probability of an assault occurring in the next 3 days” SHAP tells us which features drove that prediction upward or downward.

For our assault prediction model (representative of the approach across all four crime types), we engineered the following 18 features:

- **neighbourhood_enc** (encoded neighbourhood)
- **week_day_num** (day of week, 0-6)
- **season_num** (season, 0-3)
- **holiday** (holiday flag, 0/1)
- **hist_rate_3w** (historical assault rate, 3 weeks)
- **hist_rate_6w** (historical assault rate, 6 weeks)
- **periods_since_last** (periods since last assault)
- **month_sin** (monthly cyclical pattern, sine)
- **month_cos** (monthly cyclical pattern, cosine)
- **period_sin** (weekly cyclical pattern, sine)
- **period_cos** (weekly cyclical pattern, cosine)

- `ewma_3w` (exponentially weighted moving average, 3-week)
- `rate_change_3w` (rate of change: 3-week minus 6-week)
- `assault_trend` (local trend slope)
- `recent_assault_density` (recent assault concentration)
- `seasonal_pattern` (year-over-year seasonal component)
- `year_continuous` (continuous year for long-term trend)
- `global_yoy_change` (city-wide year-over-year change)

These features collectively capture temporal patterns, historical rates, trend dynamics, cyclical behaviors, and neighbourhood-specific baselines.

When we applied SHAP analysis to our trained models, a clear pattern emerged: the most influential features were consistently the temporal and historical trend features (`hist_rate_3w`, `ewma_3w`, `periods_since_last`, and `recent_assault_density`), while static neighbourhood encoding played a comparatively smaller role. This finding has important implications: the model relies heavily on recent historical patterns and momentum rather than stable neighbourhood characteristics.

2.4.2 Critical Limitation: Temporal Dependency and Model Decay

The SHAP analysis reveals a fundamental limitation of our current approach. Because predictions depend so strongly on recency-weighted historical trends and temporal patterns, the model’s accuracy will degrade over time if it is not continuously updated. Specifically:

- As we move further from the training period (2014-2024), the historical trend features become less representative of current conditions
- The model cannot anticipate structural shifts in crime patterns (e.g., post-pandemic changes, new policing strategies, demographic shifts)
- Without retraining, the `year_continuous` and `global_yoy_change` features will extrapolate trends beyond their valid range

Therefore, a critical operational requirement is **frequent retraining**. Once 2026 and 2027 data become available, they must be incorporated into the training set to recalibrate historical rates, moving averages, and trend slopes. Without this ongoing maintenance, prediction accuracy will decline significantly, potentially rendering the model unreliable for real-world decision-making by 2027. We recommend quarterly or semi-annual retraining cycles to maintain model trustworthiness.

2.5 Web Interface Implementation

A website was implemented using Next.js for the front end and FastAPI to connect the Python back end to the front end. The website allows users to select a neighbourhood and crime type they want to predict within a time range, then provides predictions and probability of occurrence based on the model’s expectations.

3 Results

All figures supporting these results are presented in the Annex (Section 7). For both crime and traffic data, we created maps indicating the safety score of each neighbourhood, coloured from green (safest) to red (most dangerous).

By examining crime results between 2020-2025, assault offences are the most common across many years. Scatterplots show a negative correlation between the number of selected incidents and safety score, which is expected: fewer crimes yield a higher safety score, and more crimes yield a lower safety score. Assault is the highest type of offence for most neighbourhoods regardless of safety score. Auto theft and break and enter are the two most common offences for neighbourhoods with stronger safety scores. In 2020, Moss Park had the most reported cases; from 2021-2025, West Humber-Claireville had the most reported cases.

For traffic results between 2020-2025, the trendline for collision types shows a decrease in 2020 due to the COVID-19 pandemic, but a notable increase through 2022, peaking in 2024. According to the map illustration and bar chart, Wexford/Maryvale is the neighbourhood with the lowest safety score and highest collision value, making it the most dangerous neighbourhood based on traffic collisions.

Based on crime safety scores, Moss Park and West Humber-Claireville are the two most dangerous neighbourhoods for crime. For traffic safety scores, Wexford/Maryvale is the most dangerous neighbourhood for traffic. Interestingly, West Humber-Claireville appears in the top rankings for both crime and traffic, suggesting it may have the highest overall safety risk. However, our safety scores for crime and traffic do not correlate, indicating the two scores are independent when analysing neighbourhoods.

4 Limitations

We chose the values of weights used for calculating safety scores rather than validating them against an external standard. Changes in weights would alter results and produce different neighbourhood rankings. There was a limited amount of data to train the models, which means the models have less data to learn from, reflected in their lower accuracy scores for less frequent crime types. The crime and traffic safety scores were calculated separately, meaning we did not measure the correlation between the two scores. Our data only includes crimes reported to the police department, so safety scores may not fully reflect the actual level of risk in every neighbourhood.

Additionally, our predictive models lack access to several potentially valuable data sources that could enhance prediction accuracy. We did not have access to:

- **Demographic and population data:** Population density, age distribution, and residential vs. commercial land use patterns would help contextualize crime rates per capita.
- **Socioeconomic indicators:** Income levels, employment rates, education attainment, and housing affordability could reveal underlying risk factors.
- **Environmental and situational data:** Weather conditions, lighting infrastructure, presence of CCTV cameras, public transit proximity, and time of day for each incident.

- **Event-specific contextual information:** Details about instigating circumstances, relationship between parties, or whether alcohol/substances were involved.

Consequently, our current models rely primarily on heuristic temporal trends (day of week, season, holidays) and neighbourhood-level historical patterns to make predictions. While this approach yields reasonably accurate results particularly for high-frequency crime types like assault and auto theft the models remain fundamentally limited in their ability to capture causal or contextual risk factors. Future work integrating richer external datasets would likely improve both predictive performance and interpretability.

5 Next Steps

Several promising directions exist to build upon the foundation established in this project.

5.1 Refining Safety Score Weights

The weights used in our safety score calculations (CSI weights, offence weights, and severity weights for traffic collisions) were chosen based on domain reasoning rather than empirical validation. A key next step would be to run systematic simulations to optimize these coefficients. We could employ techniques such as:

- Sensitivity analysis to understand how changes in individual weights affect neighbourhood rankings
- Calibration against external benchmarks, such as public perception surveys of neighbourhood safety
- Conjoint analysis or pairwise comparison surveys where residents rank neighbourhood safety scenarios, allowing us to infer socially preferred weightings

This would transform our safety scores from researcher-determined metrics into empirically grounded, community-calibrated indices.

5.2 Geographic and Cross-Jurisdictional Expansion

While our current analysis focuses exclusively on Toronto, the methodology is designed to be portable. Future work should apply the same modeling pipeline to:

- Other Canadian municipalities (Vancouver, Montreal, Calgary) to test generalizability
- International cities with similar open data infrastructures (Chicago, London, Los Angeles)
- Cross-jurisdictional comparative analysis to identify whether the relationships we observed such as the independence of crime and traffic safety scores hold across different urban environments

Such expansion would not only validate our approach but also reveal city-specific versus universal patterns in crime and traffic safety.

5.3 Socioeconomic Analysis and Causal Inference

A primary objective from the outset has been to explore connections between crime patterns and socioeconomic conditions. While our current safety scores identify *where* risk is highest, they do not explain *why*. Next steps include:

- Integrating public census data (Statistics Canada, City of Toronto Open Data) to examine relationships between crime types and neighbourhood characteristics such as:
 - Median household income and income inequality (Gini coefficient)
 - Employment rates and occupational distribution
 - Educational attainment levels
 - Housing tenure (rent vs. own) and affordability metrics
 - Newcomer/immigrant concentration and years since arrival
- Developing a companion Power BI dashboard specifically for socioeconomic-visualisation, enabling users to overlay crime safety scores with demographic layers, revealing spatial correlations at a glance
- Exploring whether we can build predictive models that use socioeconomic indicators as features, potentially forecasting crime risk in neighbourhoods based on demographic shifts before crime patterns emerge
- Remaining objective and transparent about the limitations of such analysis, correlation does not imply causation, and socioeconomic factors are associated with crime through complex, multi-causal pathways that require careful interpretation

5.4 Data Enrichment and Model Enhancement

As noted in the Limitations section, our current models rely primarily on temporal and neighbourhood-level heuristics. To address this, we plan to:

- Source richer datasets, including:
 - Weather data (precipitation, temperature, visibility) to capture environmental influences on crime and collision occurrence
 - Point-of-interest data (locations of bars, schools, transit stations, nightlife venues)
 - Lighting infrastructure and 311 service requests for streetlight maintenance
 - Temporal fine-grained data (hour-level occurrence times) currently not consistently available in the open dataset
- Revisit and augment the feature engineering pipeline with these new data sources
- Explore alternative modeling architectures beyond LightGBM, particularly:
 - **LSTMs and GRUs:** To better capture sequential dependencies in crime patterns across time

- **Spatio-temporal Graph Neural Networks:** To model how crime risk propagates between neighbouring areas
- **Transformer-based models:** For attention over both temporal and spatial dimensions
- Conduct rigorous A/B testing of model improvements using held-out validation periods

5.5 Community-Facing Applications

Finally, we aim to translate this work into practical tools for Toronto residents and community safety organizations:

- Develop automated alerting systems for significant predicted increases in specific crime types
- Partner with neighbourhood community groups to validate findings and incorporate local knowledge

Collectively, these next steps would transform our current retrospective analysis into a proactive, predictive, and community-engaged safety platform.

6 References

- Xu Zhang, Lin Liu, Minxuan Lan, Guangwen Song, Luzi Xiao, Jianguo Chen, Interpretable machine learning models for crime prediction, *Computers, Environment and Urban Systems*, Volume 94, 2022, 101789, ISSN 0198-9715, <https://www.sciencedirect.com/science/article/abs/pii/S0198971522000333>
- <https://dl.acm.org/doi/epdf/10.1145/3797161.3797199>
- Nicho, M., Hamed, A., Gaber, T., & Al Arimi, J. H. (2025). A-XGBoost: a resilient machine learning technique for predicting crimes against women across cultures on low cardinality crime data. *Cogent Social Sciences*, 11(1). <https://doi.org/10.1080/23311886.2025.2527392>
- GeeksForGeeks, (2021). <https://www.geeksforgeeks.org/machine-learning/lightgbm-vs-xgboost-which-algorithm-is-better/>
- Toronto Open Data, Major Crime Indicators, <https://open.toronto.ca/dataset/major-crime-indicators/>
- Toronto Open Data, Motor Vehicle Collisions involving Killed or Seriously Injured Persons, <https://open.toronto.ca/dataset/motor-vehicle-collisions-involving-killed-or-seriously-injured-persons/>
- Toronto Police Service, Open Collision Data, <https://data.tps.ca/maps/bc4c72a793014a55a674984ef175a6f3>
- Toronto Police Service, General Community Safety Indicators, https://data.tps.ca/datasets/0a239a5563a344a3bbf8452504ed8d68_0/explore

- Toronto Police Service, Shooting and Firearm Discharge, https://data.tps.ca/datasets/64ddeca12da34403869968ec725e23c4_0/explore
- Toronto Police Service, Homicide, https://data.tps.ca/datasets/d96bf5b67c1c49879f354dad51cf81f9_0/explore
- Toronto Police Service, Assault, https://data.tps.ca/datasets/b4d0398d37eb4aa184065ed625ddb922_0/explore
- Toronto Police Service, Auto Theft, https://data.tps.ca/datasets/95ab41aee16847dba8453bf1688249d6_0/explore
- Toronto Police Service, Bicycle Theft, https://data.tps.ca/datasets/a89d10d5e28444ceb0c8d1d4c0ee39cc_0/explore
- Toronto Police Service, Break and Entering, https://data.tps.ca/datasets/040ead448df2412da252cfbb532e77ac_0/explore
- Toronto Police Service, Robbery, https://data.tps.ca/datasets/d0e1e98de5f945faa2fe635dee3f4062_0/explore
- Toronto Police Service, Theft Over, https://data.tps.ca/datasets/7530d9b637c340059ccb81a782481c04_0/explore
- Toronto Police Service, Theft From Motor Vehicle, https://data.tps.ca/datasets/d9303bc20f8a4351b7744a8703eecb80_0/explore

A Appendix: Images

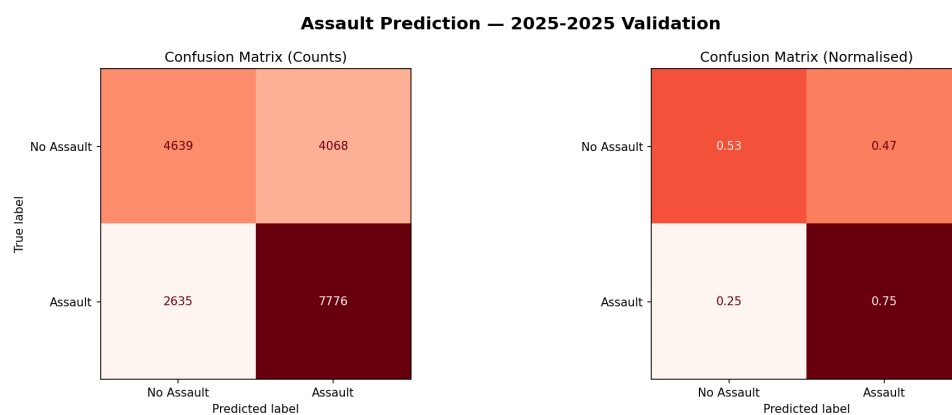


Figure 1: Confusion Matrix - Assault Model

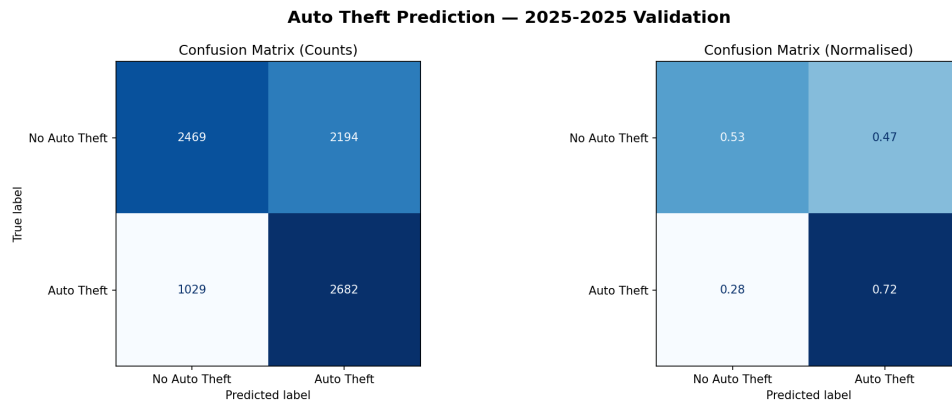


Figure 2: Confusion Matrix - Auto Theft Model

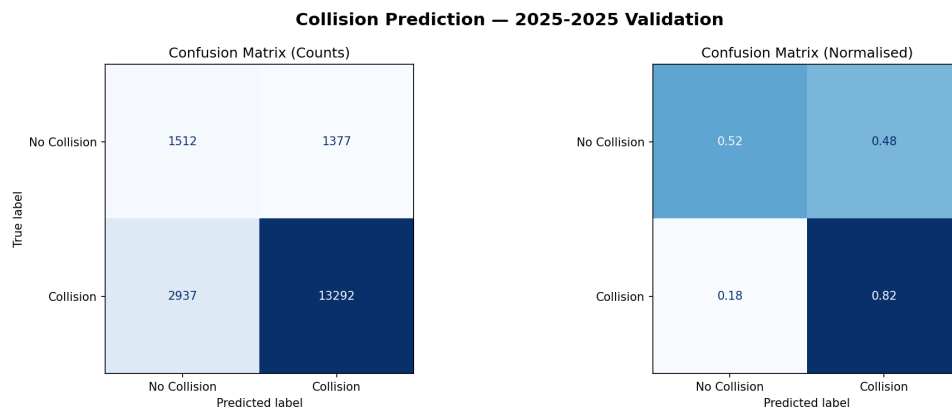


Figure 3: Confusion Matrix - Collision Model

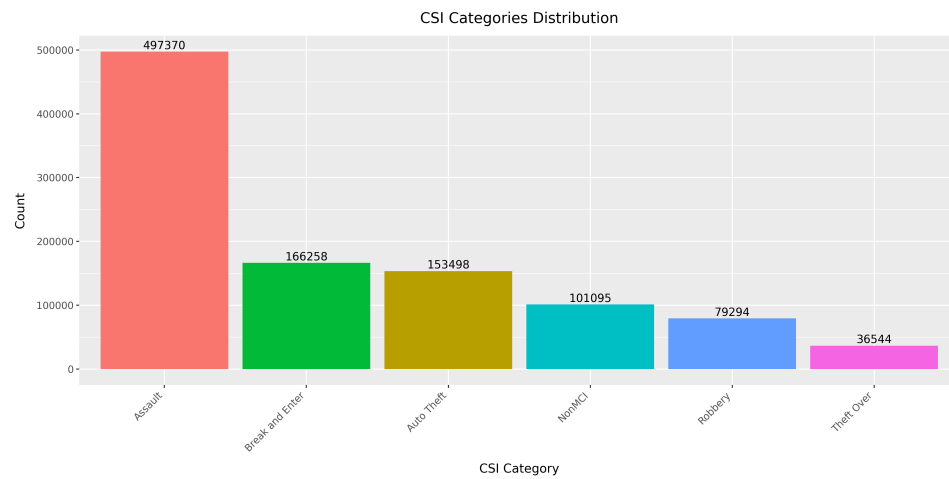


Figure 4: CSI Categories Distribution

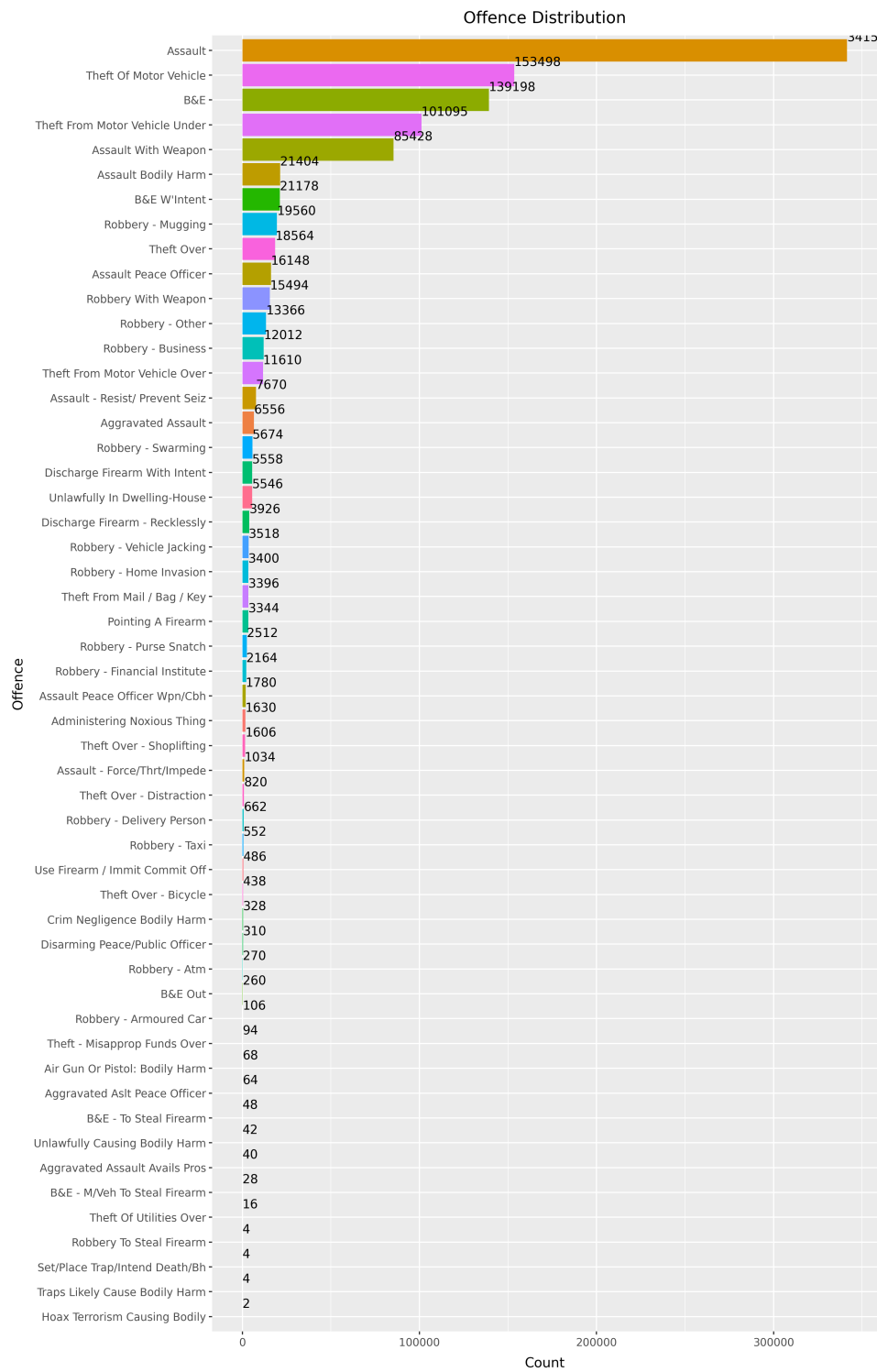


Figure 5: Offences Distribution

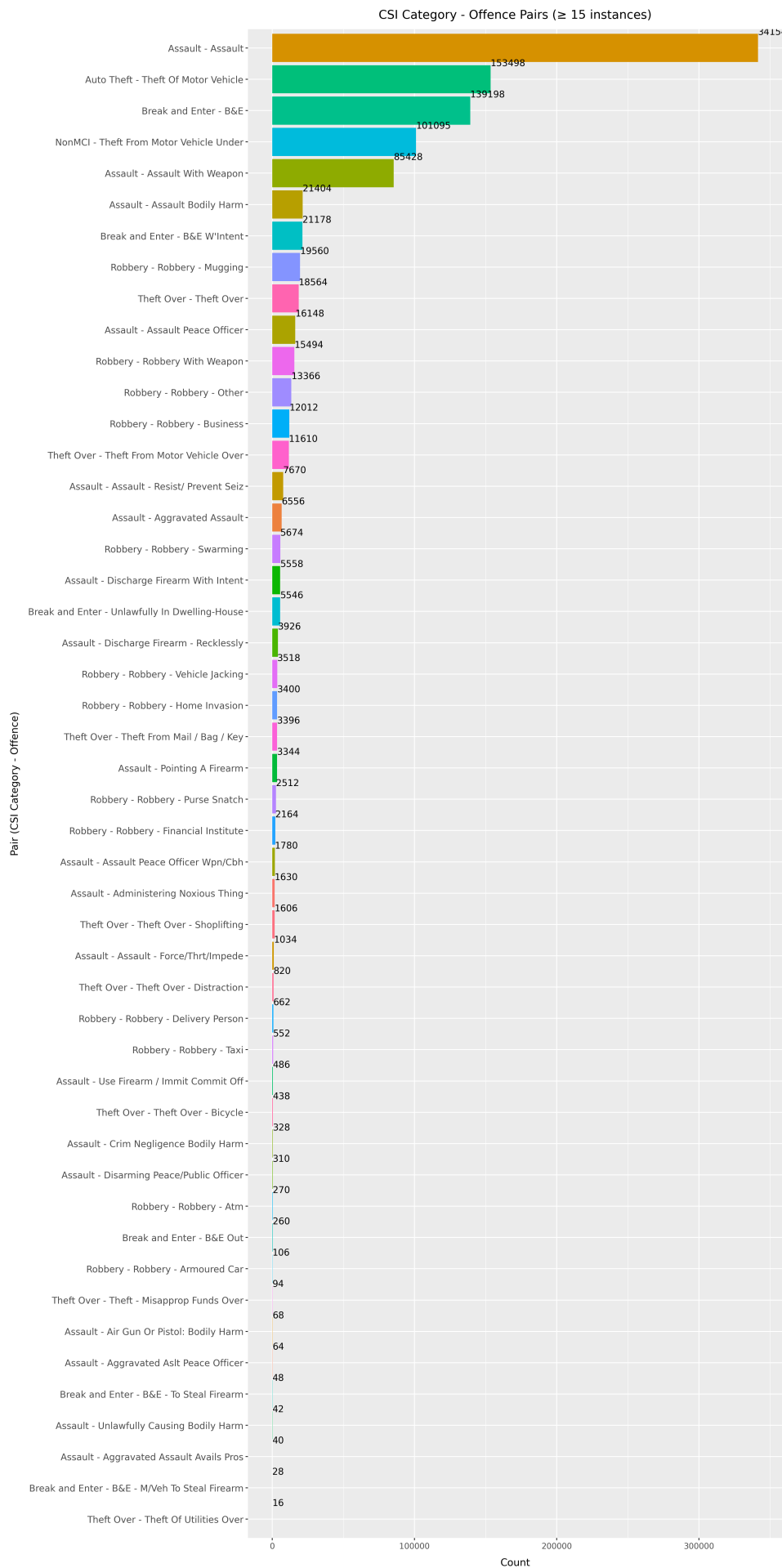
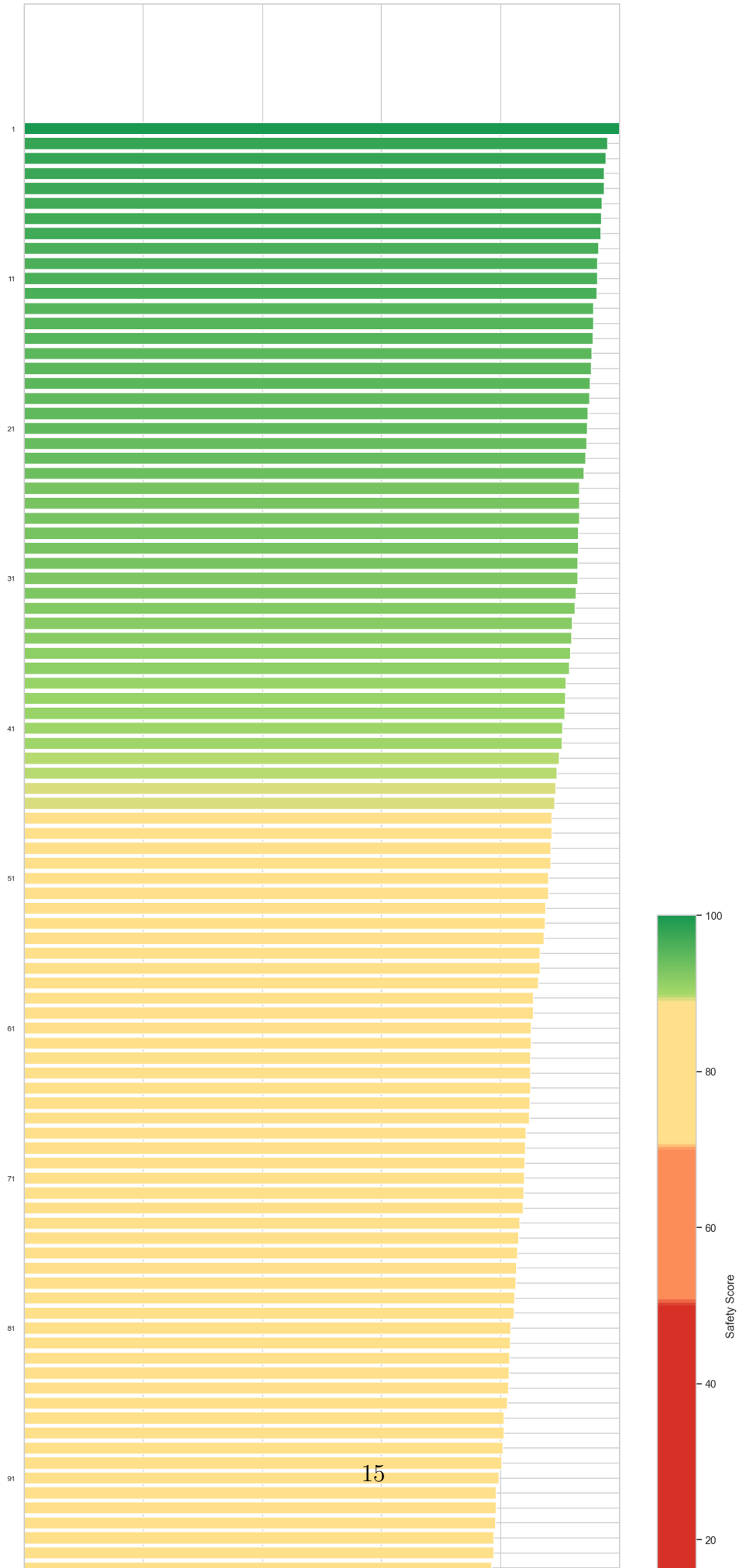


Figure 6: Pair Frequencies Analysis

All Neighbourhoods — Safety Score Heatmap



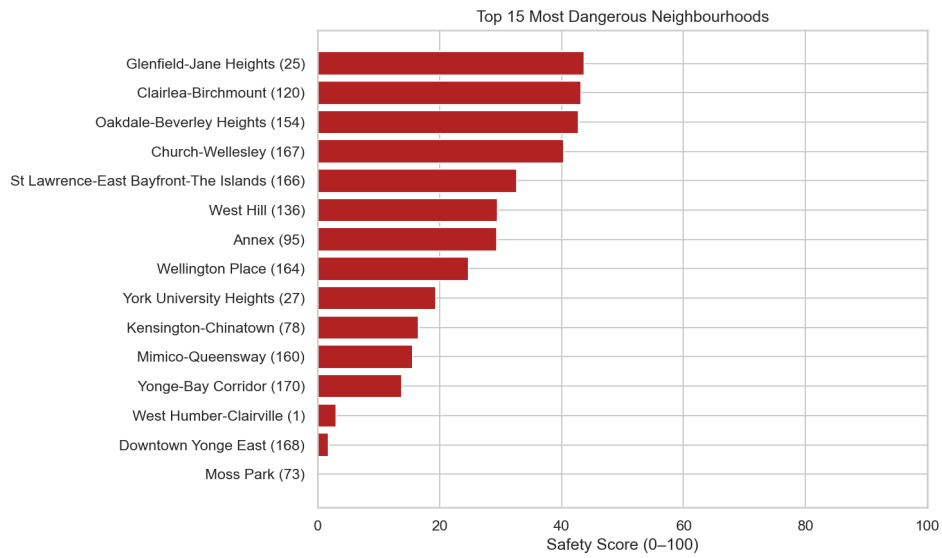


Figure 8: Top 15 Most Dangerous Neighbourhoods

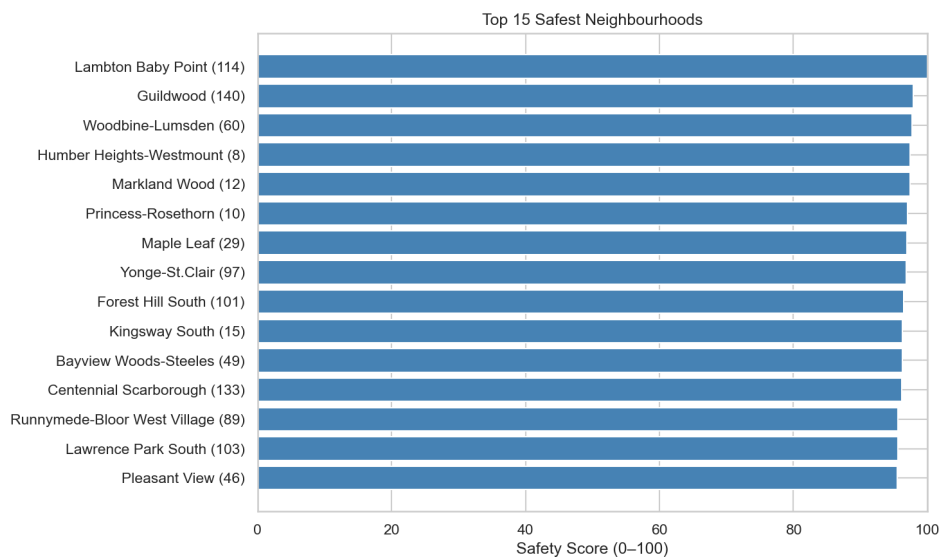


Figure 9: Top 15 Safest Neighbourhoods

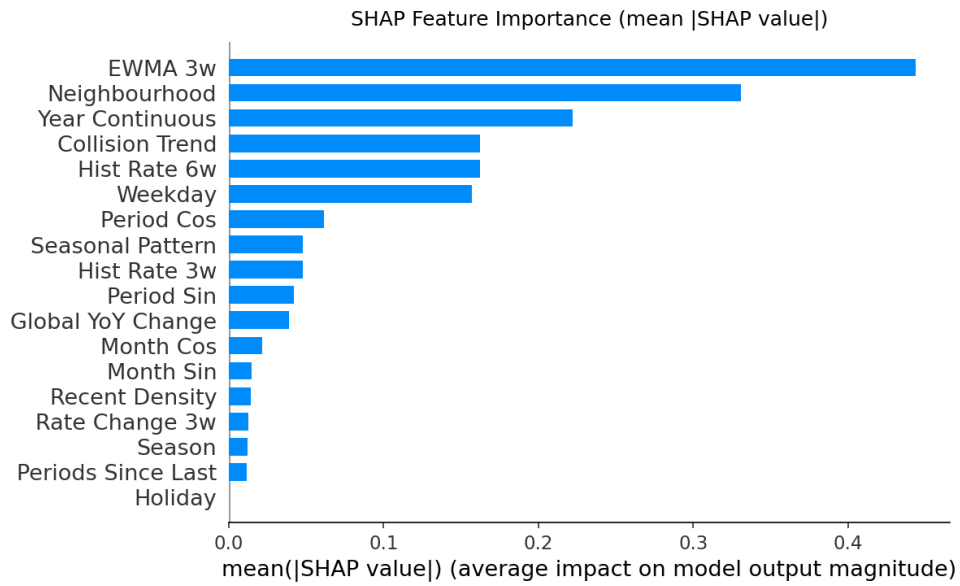


Figure 10: SHAP Bar Plot - Collision Model

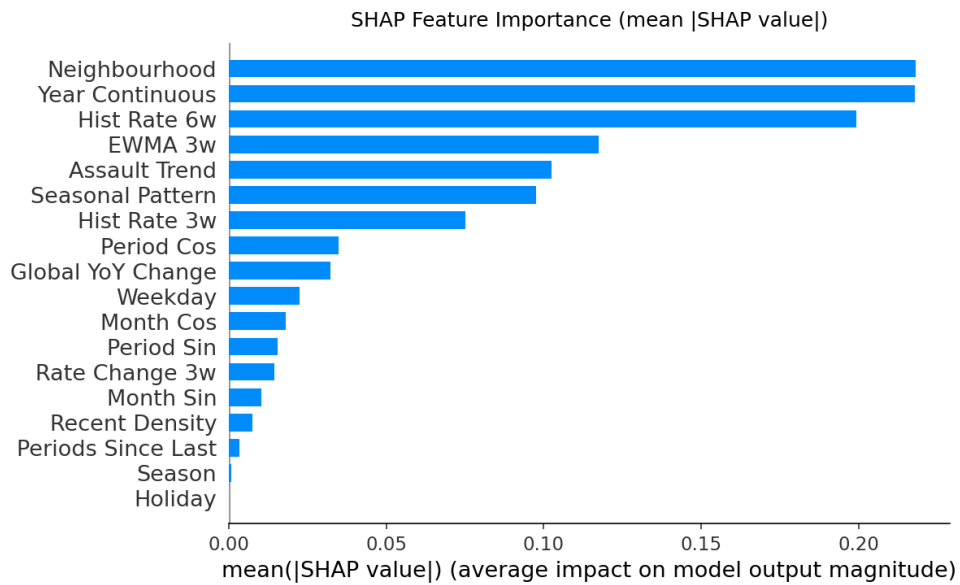


Figure 11: SHAP Bar Plot - Assault Model

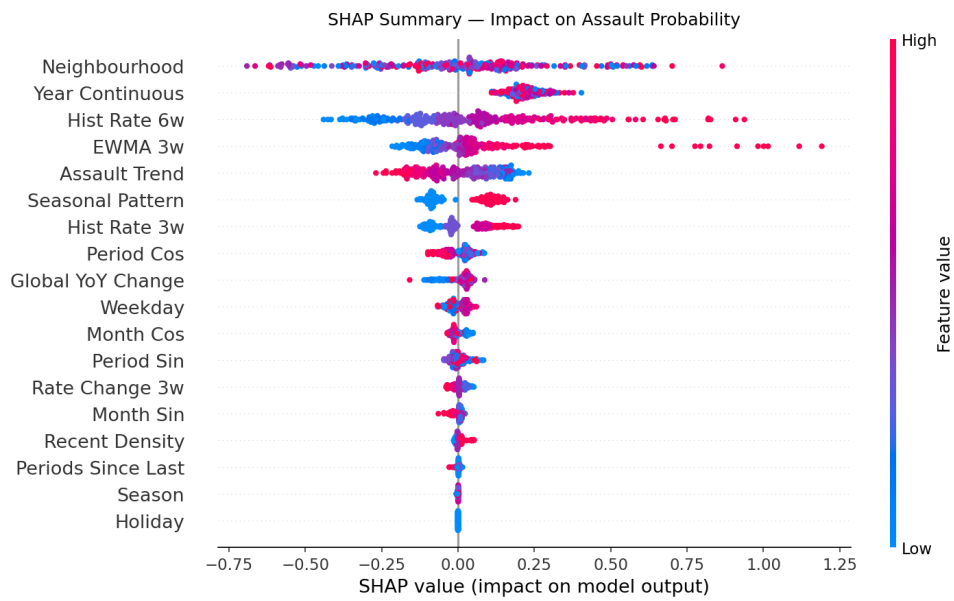


Figure 12: SHAP Summary Plot - Assault Model

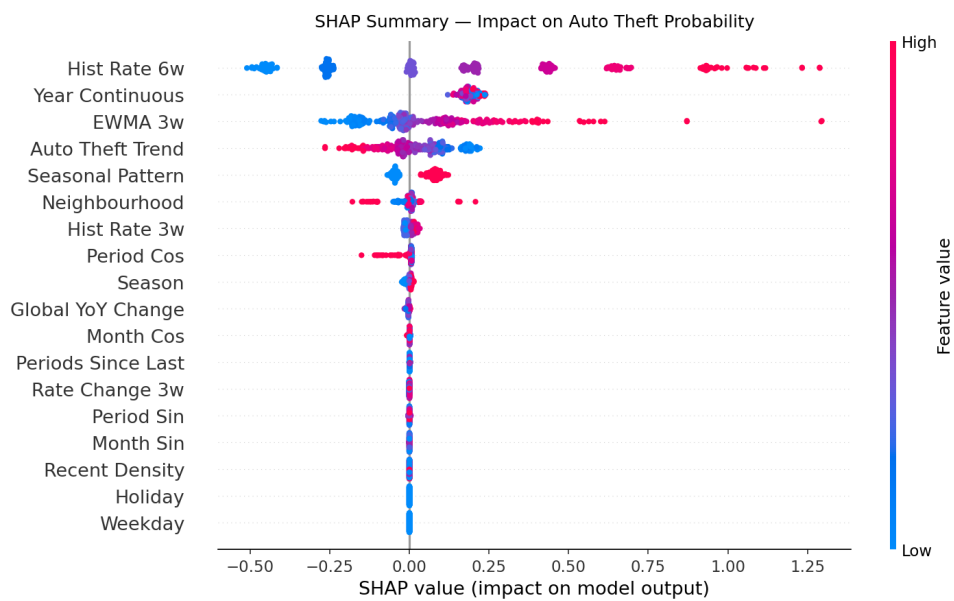


Figure 13: SHAP Summary Plot - Auto Theft Model

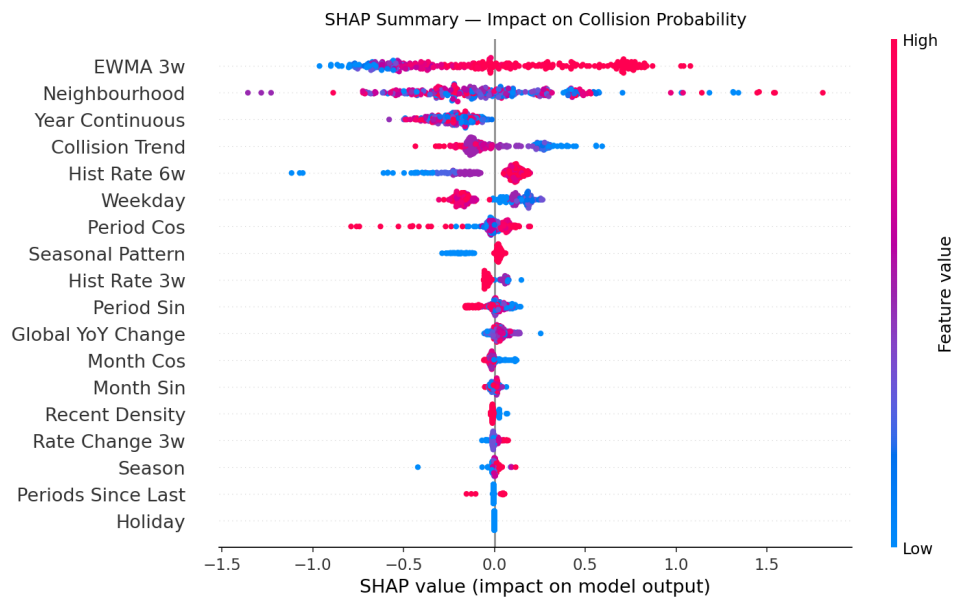


Figure 14: SHAP Summary Plot - Collision Model

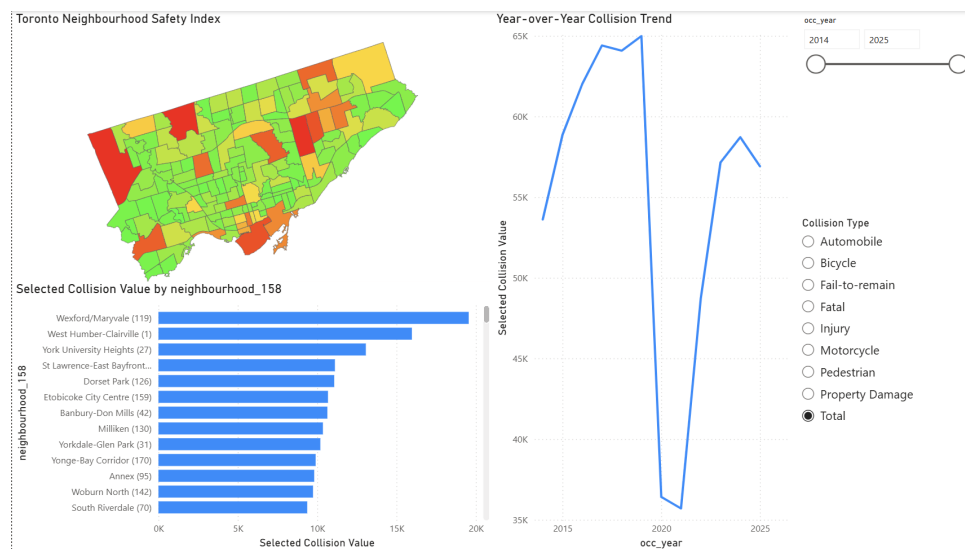


Figure 15: Power BI Collision Dashboard - View 1

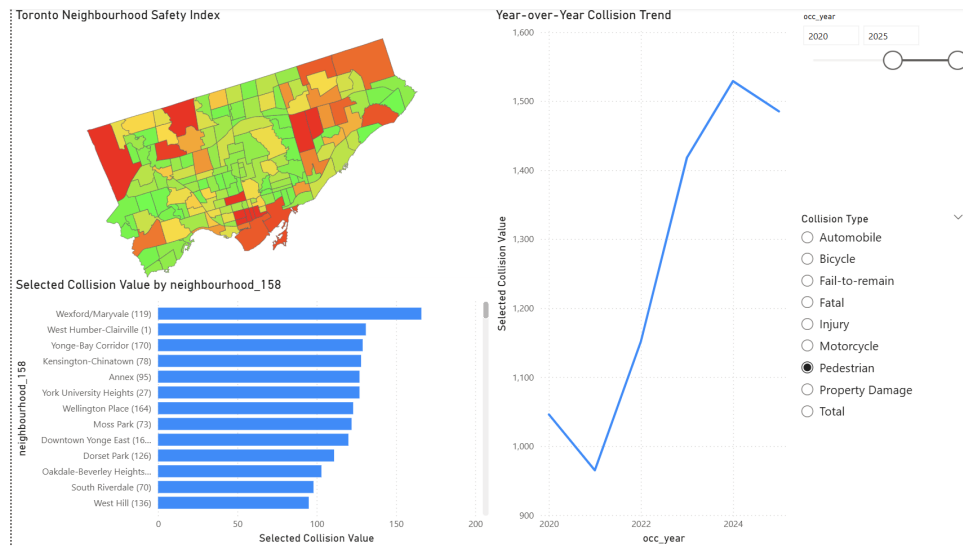


Figure 16: Power BI Collision Dashboard - View 2

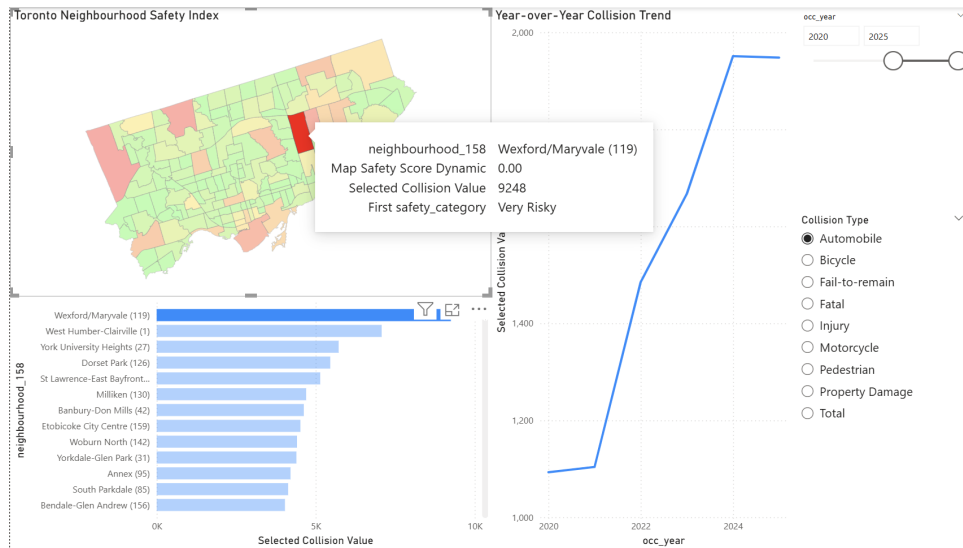


Figure 17: Power BI Collision Dashboard - View 3

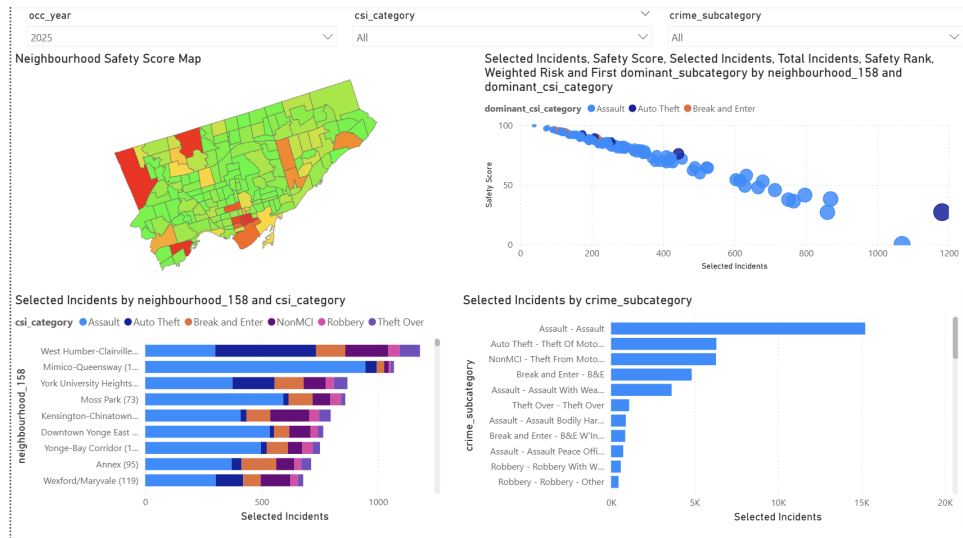


Figure 18: Power BI Crime Dashboard - View 1

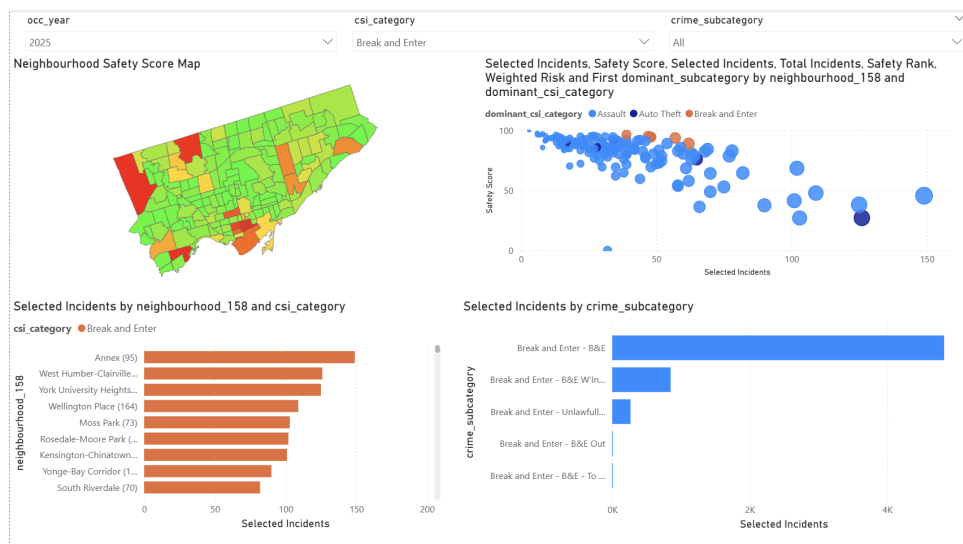


Figure 19: Power BI Crime Dashboard - View 2

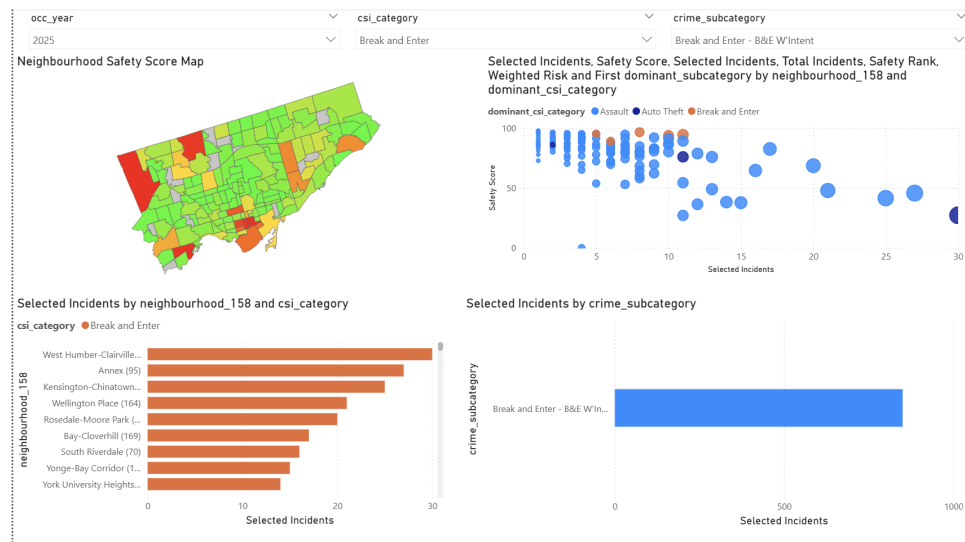


Figure 20: Power BI Crime Dashboard - View 3